

## **DEMO AKAN GRAMMAR**

The grammar demonstrates a small Lexical-Functional Grammar (LFG) fragment for Akan, a Kwa language spoken in Ghana. The aim is to model three phenomena in Akan: clitics, tense/aspect/mood (TAM) marking, and serial verb constructions.

Developing the grammar I realised some of these phenomena are less straightforward than they initially appeared and as someone who does not speak the language as fluently, this was a big learning curve, not only of LFG but of Akan grammar. For example, that the subject-agreement prefix is better analysed as a cliticized form of an existing free pronoun rather than as generic agreement. Also how interaction between different parts of the grammar also became important, like the ATR value of a verb stem determining which past-tense suffix is grammatical, meaning that the analysis of vowel harmony has consequences for the tense system built on top of it. Some features were introduced but not implemented only because I hope to expand upon the grammar later.

### **1. Cliticization**

Akan has free-standing subject pronouns, but when a pronoun occurs immediately before the verb, it obligatorily cliticizes to the verb and appears as a bound prefix rather than remaining a separate word. This cliticization is specific to the subject position. A full lexical subject NP and the cliticized pronoun are therefore in complementary distribution rather than being freely interchangeable. A full NP subject and the clitic cannot both occupy the subject position at the same time.

The grammar uses the simpler of two possible designs by making the subject NP directly optional in the S rule rather than introducing a separate subject category. When the NP is present, it supplies the SUBJ PRED information, together with features such as PERS and NUM. When it is absent, the S rule itself contributes nothing to SUBJ. The missing information instead comes from the Pref clitic on the verb. Each Pref lexical entry specifies SUBJ PRED = 'pro' together with the relevant PERS, NUM, and ANIM features. Since the verb's f-structure is passed directly

up to S through the equation (VP:  $\wedge=!$ ), these features reach SUBJ whether or not the subject NP is present.

Therefore, in an ordinary declarative clause, the absence of a full subject NP does not leave SUBJ incomplete: the subject clitic supplies the relevant information. The IMPERATIVE template provides the other source of subject information, setting SUBJ PRED = 'pro', PERS = 2, and ANIM = + directly, without requiring a Pref clitic.

## 2. Tense, Aspect, and Mood marking

TAM marking is divided between a prefix position and a past-tense suffix. The prefixes include  $b\epsilon$ -/be- for future, re- for progressive, and a- for perfect, while past tense is marked with either -ee or - $\epsilon\epsilon$  depending on vowel harmony. A complication arises in serial verb constructions, where the entire chain is treated as a single clause. This means that the verbs cannot each introduce an independent tense or aspect. Instead, the chain receives one overall TAM specification, and morphology on a non-initial verb is interpreted in relation to the TAM established by the first verb.

TENSE, ASPECT, and MOOD are represented using parameterised templates. For example,  $ASPECT(A) = (\wedge ASPECT) = A$ . Ordinary prefixes such as  $b\epsilon$ - can therefore invoke @FUTURE and @INDICATIVE. The more complicated cases are the perfect prefix a- and the past-tense suffixes, whose behaviour depends on whether the verb occurs inside a serial construction.

To capture this distinction, the VP rule marks a verb embedded in a serial chain with  $(\wedge CONTEXT) = serial$ . The relevant TAM entries then check this feature using constraining equations ( $=c$ ). Outside a serial construction, a- contributes its normal perfect-tense information. Inside a serial construction, however, it does not introduce a new TAM value. Instead, it checks that the TAM value already established for the chain is compatible with the interpretation of the non-initial verb. This prevents the morphology on the second verb from creating a separate tense or aspect specification.

The past-tense suffixes work in a similar way. Outside a serial context, -ee and - $\epsilon\epsilon$  contribute TENSE = past. Inside a serial context, they do not independently establish the tense of the second verb; they instead record the presence of the past marker while the tense of the chain comes from the appropriate source. The difference between -ee and - $\epsilon\epsilon$  is treated as purely phonological, so they are represented as separate lexical entries with identical f-structure content.

The choice between them is determined by vowel harmony rather than by a difference in syntactic or semantic features.

### 3. Serial verb constructions

A serial verb construction combines two or more verbs in a single clause without a coordinating connective. A simplified example would be “X take Y go Z,” roughly meaning that X takes Y to Z. The verbs behave as parts of one clause rather than as two independently coordinated clauses. They share a subject, will often share an object, and take only one overall TAM specification. Because of this, coordinating two independent VPs would not capture the relevant structure. Such an analysis would allow each VP to introduce its own subject, object, and TAM features. Instead, the VP rule allows one VP to contain another VP, with equations that explicitly link the two structures.

The equations  $(! \text{ SUBJ}) = (^ \text{ SUBJ})$  and  $(! \text{ OBJ}) = (^ \text{ OBJ})$  force the embedded verb's subject and object to be identical to those of the outer verb. The disjunction  $\{ (! \text{ ASPECT}) (! \text{ ASPECT}) = (^ \text{ ASPECT}) \mid (! \text{ TENSE}) (! \text{ TENSE}) = (^ \text{ TENSE}) \}$  requires the verbs to share tense or aspect rather than each asserting an independent value. Each branch also requires the embedded verb to actually carry that feature locally before it can be shared upward. Without this check, a branch with nothing to share was still vacuously satisfiable, which produced spurious extra parses for sentences where only one of ASPECT or TENSE was ever set

The equation  $(! \text{ CONTEXT}) = \text{serial}$  connects this recursive VP structure to the lexical TAM entries described above. Once the embedded verb is marked as belonging to a serial context, its TAM morphology can check the already established TAM value instead of asserting a new one. The set-membership condition  $! \$ (^ \text{ ADJUNCT})$  also prevents the embedded VP from being analysed as an ordinary post-verbal adjunct.

Finally, the trailing NP rule,  $(\text{NP}: (! \text{ OBJ}) = !)$ , allows the second verb to introduce its own object when the two verbs do not share one. This covers chains such as “take knife kill chicken,” where each verb has a different object but both verbs still share the same subject. The resulting analysis therefore captures both shared-argument serial constructions and cases in which the verbs have separate objects, while maintaining the analysis of the sequence as a single clause.

## **APPENDIX**

### **Positive Sentences**

1. Me- kɔ  
1SG- go  
'I go'
2. ɛ- hiri  
3SG.INAN- jumps  
"It jumps"
3. Kofi kɔ dan no akyi  
Kofi go building DET behind  
"Kofi is gone behind the building"
4. Akokɔ no hiri  
Chicken DET jump  
"The chicken jumps"
5. ɔ- re-kan sika no  
3SG- PROG-count money DET  
"He is counting the money"
6. Me- pɛ no  
1SG-like 3SG.OBJ  
"I like him/her"
7. Me- pɛ nwoma no  
1SG-like book DET  
"I like the book"
8. Gyae  
Stop  
"Stop"
9. Sɔre  
Rise  
"Get up"

10. Me- ma Kofi sika  
1SG- give Kofi money  
“I give Kofi money”
11. Kofi kum akokɔ no  
Kofi kill chicken DET  
“Kofi kills chicken the”
12. Ɔ- kum akokɔ no  
3SG-kill chicken DET  
“He/ she kills the chicken”
13. Wo kum akokɔ no noa -εε  
3PL-kill chicken DET cook -PST  
“They killed the chicken and cooked it”
14. Ɔ- bε- hiri  
3SG-FUT jump  
“He/ she will jump”
15. Yε- kɔ -εε  
1PL-go-past  
“We went”

### Negative Sentences

16. Me- no pε  
1SG- 3SG.OBJ like  
“I it/he/she like”
17. Me- pε kɔ  
1SG-like go  
“Me like go”
18. Ɔ- kum  
3SG-kill  
“He/she kills”

19. Anoma bi ε- da dua no ase

Bird INDEF 3SG.INAN- sleep tree DET under

“A bird is sleeping under the tree”

20. Dua ε- hiri

Tree 3SG.INAN- jump

“The tree jumps”